1. Spring 2024 Final Term Linear Algebra Model Solution

# University of Central Punjab
## Faculty of Information Technology
### Final Term Exam

| Program | BSCS | Semester | Spring 2024 |
|---|---|---|---|
| Course Title | Linear Algebra | Course Code | CSSS2753 |
| Course Coordinator | Dr. Maria Naseem | Section | |
| Time Allowed | 2 hour 30 minutes | Total Marks | |
| Date | 24-07-2024 | Time Slot | 9:00 AM - 11:30 AM |
| Student Name | | Registration No. | |

**Instructions**

1. Write your name and registration number on each page of the answer sheet and sign those answer sheets before submission.
2. Attempt all questions.
3. All your calculations should be on your answer sheet.
4. You have first 5 minutes to read the paper carefully and then start solving the paper.

**Q1. [Marks: $5 + 1 + 4 = 10$]**

A robotics company is analysing the stability of three connected mechanical arms in a system. The interactions and counter-forces between these arms are represented by a 3x3 matrix $M$, where each entry $m_{ij}$ represents the force exerted by arm $i$ on arm $j$. A negative value indicates a counteracting force. The matrix M is given by:

$$M = \begin{bmatrix} 11 & -8 & 4 \\ -8 & -1 & -2 \\ 4 & -2 & -4 \end{bmatrix}$$

a) Compute the eigenvalues of the matrix $M$.

b) Interpret the meaning of these eigenvalues in the context of the mechanical arms' stability.

c) Find the corresponding eigenvector for any one of the eigenvalues.

**Q2. [Marks: $3 + 3 + 3 + 1 = 10$]**

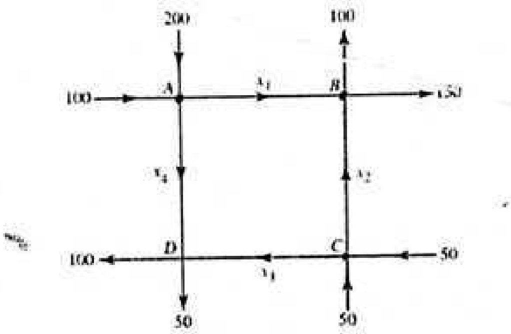
Consider the following vectors from $\mathbb{R}^3$.

$$S = \{\vec{u}_1 = (1,1,0), \vec{u}_2 = (0,2,3), \vec{u}_3 = (1,2,3), \vec{u}_4 = (3,6,6)\}$$

a) Determine whether the vectors in set $S$ are linearly independent or linearly dependent?
b) If the vectors in set $S$ are linearly dependent, is it possible to write one vector as a linear combination of others?
c) Determine whether the set $S$ spans the $\mathbb{R}^3$ space. Justify your answer.
d) Can $S$ be basis for the $\mathbb{R}^3$ space. Justify your answer.

**Q3. [Marks: $2 + 5 + 2 + 1 = 10$]**

Construct a mathematical model that describes the traffic flow in the road network depicted in figure below. All streets are one-way streets in the direction indicated. The units are vehicles per hour.




a) Write down the linear system that models the network flow problem.
b) Solve the linear system by applying Gauss Jordan Method to determine the solution.
c) Give two distinct possible flows of traffic.
d) What is the minimum possible flow that can be expected along branch AB.

**Q4. [Marks: $3 + 5 + 2 = 10$]**

Consider the following equation of circle

$$(x+2)^2 + (y-1)^2 = 4$$

$A = \begin{bmatrix} 1 & 0 \\ 0 & k \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$

a) Translate this circle by vector $(2,3)$.
b) Find equation of translated circle under the effect of stretching parallel to y-axis by factor 3.
c) Sketch the original and the images of circle obtained in part 'a' and 'b'.

**Q5. [Marks: 7+3=10]**

A message is converted into numeric form by letting

| | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26=0 |

a) Encode the message "SECRET WORLD" with key matrix

$$K = \begin{bmatrix} 5 & 1 \\ 2 & 7 \end{bmatrix}$$

b) Find the inverse of $K$ under modulo 26.

Q1:

$$M = \begin{bmatrix} 11 & -8 & 4 \\ -8 & -1 & -2 \\ 4 & -2 & -4 \end{bmatrix}$$

(a) Compute the eigen values of the matrix M.

$$M - \lambda I = \begin{bmatrix} 11-\lambda & -8 & 4 \\ -8 & -1-\lambda & -2 \\ 4 & -2 & -4-\lambda \end{bmatrix} \text{——} ①$$

put $\det(M - \lambda I) = 0$

$$\begin{vmatrix} 11-\lambda & -8 & 4 \\ -8 & -1-\lambda & -2 \\ 4 & -2 & -4-\lambda \end{vmatrix} = 0$$

Expanding from $R_1$, we have

$$(11-\lambda)\begin{vmatrix} -1-\lambda & -2 \\ -2 & -4-\lambda \end{vmatrix} - (-8)\begin{vmatrix} -8 & -2 \\ 4 & -4-\lambda \end{vmatrix} + 4\begin{vmatrix} -8 & -1-\lambda \\ 4 & -2 \end{vmatrix} = 0$$

$$(11-\lambda)\left[(-1-\lambda)(-4-\lambda)-(-2)(-2)\right]+8\left[(-8)(-4-\lambda)-(-2)(4)\right]$$

$$+4\left[(-8)(-2)-(4)(-1-\lambda)\right]=0$$

$$(11-\lambda)\left[\cancel{4}+\lambda+4\lambda+\lambda^2-\cancel{4}\right]+8\left[32+8\lambda+8\right]$$

$$+4\left[16+4+4\lambda\right]=0$$

$$(11-\lambda)\left[\lambda^2+4\lambda+\lambda\right]+8\left[40+8\lambda\right]+4\left[20+4\lambda\right]=0$$

$$(11-\lambda)\left[\lambda^2+5\lambda\right]+8\left[40+8\lambda\right]+4\left[20+4\lambda\right]=0$$

$$11\lambda^2+55\lambda-\lambda^3-5\lambda^2+320+64\lambda+80+16\lambda=0$$

$$-\lambda^3+11\lambda^2-5\lambda^2+55\lambda+64\lambda+16\lambda+320+80=0$$

$$-\lambda^3+6\lambda^2+135\lambda+400=0$$

$$\lambda^3-6\lambda^2-135\lambda-400=0 \quad \text{——} \quad ②$$

$\lambda=-5$ is the root of equation ②, So by Synthetic division

$$\begin{array}{r|rrrr} -5 & 1 & -6 & -135 & -400 \\ & \downarrow & -5 & 55 & 400 \\ \hline & 1 & -11 & -80 & \boxed{0=R} \end{array}$$

Hence $\lambda = -5$ is the root.

Now

$$\lambda^2 - 11\lambda - 80 = 0$$

$$\lambda^2 - 16\lambda + 5\lambda - 80 = 0$$

$$\lambda(\lambda-16) + 5(\lambda-16) = 0$$

$$(\lambda-16)(\lambda+5) = 0$$

i.e

$$\lambda - 16 = 0 \quad , \quad \lambda + 5 = 0$$

$$\boxed{\lambda = 16} \quad , \quad \boxed{\lambda = -5}$$

Hence

$\lambda_1 = -5$ , $\lambda_2 = -5$ , $\lambda_3 = 16$ are

the eigen values of matrix M.

Q1 (c):

$$(M-\lambda I)\vec{x} = \vec{0}$$

$$\begin{bmatrix} 11-\lambda & -8 & 4 \\ -8 & -1-\lambda & -2 \\ 4 & -2 & -4-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

For eigen value $\lambda = -5$, we have

$$\begin{bmatrix} 11+5 & -8 & 4 \\ -8 & -1+5 & -2 \\ 4 & -2 & -4+5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 16 & -8 & 4 \\ -8 & 4 & -2 \\ 4 & -2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$16x - 8y + 4z = 0$$
$$-8x + 4y - 2z = 0$$
$$4x - 2y + z = 0$$

In matrix form

$$\left[\begin{array}{ccc|c} 16 & -8 & 4 & 0 \\ -8 & 4 & -2 & 0 \\ 4 & -2 & 1 & 0 \end{array}\right]$$

$R_1 \times \left(\frac{1}{4}\right)$ , $R_2 \times \left(-\frac{1}{2}\right)$

$$\left[\begin{array}{ccc|c} 4 & -2 & 1 & 0 \\ 4 & -2 & 1 & 0 \\ 4 & -2 & 1 & 0 \end{array}\right]$$

This implies that

$$4x - 2y + z = 0$$

let $z = t$ , $y = s$

$$4x - 2s + t = 0$$

$$4x = 2s - t$$

$$x = \frac{1}{4}(2s - t) = \frac{2}{4}s - \frac{1}{4}t$$

$$\boxed{x = \frac{1}{2}s - \frac{1}{4}t}$$

$$\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{1}{2}s - \frac{1}{4}t \\ s \\ t \end{bmatrix}$$

$$\vec{x} = \begin{bmatrix} \frac{1}{2}s \\ s \\ 0 \end{bmatrix} + \begin{bmatrix} -\frac{1}{4}t \\ 0 \\ t \end{bmatrix}$$

$$\vec{x} = s \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -\frac{1}{4} \\ 0 \\ 1 \end{bmatrix}$$

Hence the eigen vector corresponding to the eigen value $\lambda = -5$ are, as below:

$$\begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} -1/4 \\ 0 \\ 1 \end{bmatrix}.$$

#2

$S = \{ \vec{u}_1 = (1,1,0) , \vec{u}_2 = (0,2,3) , \vec{u}_3 = (1,2,3) , \vec{u}_4 = (3,6,6) \}$

Sol:

(a) Linearly Independent / Linearly dependent

$$k_1 u_1 + k_2 \vec{u}_2 + k_3 \vec{u}_3 + k_4 \vec{u}_4 = \vec{0}$$

$$k_1(1,1,0) + k_2(0,2,3) + k_3(1,2,3) + k_4(3,6,6) = (0,0,0)$$

$$(k_1, k_1, 0) + (0, 2k_2, 3k_2) + (k_3, 2k_3, 3k_3) + (3k_4, 6k_4, 6k_4) = (0,0,0)$$

$$(k_1 + k_3 + 3k_4 , k_1 + 2k_2 + 2k_3 + 6k_4 , 3k_2 + 3k_3 + 6k_4) = (0,0,0)$$

Then system of equation is

$$k_1 + k_3 + 3k_4 = 0$$
$$k_1 + 2k_2 + 2k_3 + 6k_4 = 0$$
$$3k_2 + 3k_3 + 6k_4 = 0$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & 0 \\ 1 & 2 & 2 & 6 & 0 \\ 0 & 3 & 3 & 6 & 0 \end{array}\right]$$

$R_2 - R_1$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & 0 \\ 0 & 2 & 1 & 3 & 0 \\ 0 & 3 & 3 & 6 & 0 \end{array}\right]$$

$-R_2 + R_3 \quad , \quad R_3 \times (\frac{1}{3})$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 2 & 0 \end{array}\right]$$

$R_3 - R_2$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & -1 & -1 & 0 \end{array}\right]$$

$R_3 \times (-1)$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & 0 \\ 0 & 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array}\right]$$

$R_1 - R_3 \quad , \quad R_2 - 2R_3$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 2 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{array}\right]$$

By back substitution

$$K_3 + K_4 = 0$$

Let $K_4 = t$

$$\Rightarrow K_3 + t = 0$$

$$\boxed{K_3 = -t}$$

$$K_2 + K_4 = 0$$

$$\Rightarrow K_2 + t = 0$$

$$\Rightarrow \boxed{K_2 = -t}$$

$$K_1 + 2K_4 = 0$$

$$K_1 + 2t = 0$$

$$\boxed{K_1 = -2t}$$

Hence

$$\begin{bmatrix} K_1 \\ K_2 \\ K_3 \\ K_4 \end{bmatrix} = \begin{bmatrix} -2t \\ -t \\ -t \\ t \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ -1 \\ 1 \end{bmatrix} t$$

Hence, the given vectors $\vec{u}_1, \vec{u}_2, \vec{u}_3, \vec{u}_4$ are linearly dependent.

2(b)

$$\vec{u}_4 = k_1\vec{u}_1 + k_2\vec{u}_2 + k_3\vec{u}_3$$

$$(3,6,6) = k_1(1,1,0) + k_2(0,2,3) + k_3(1,2,3) \quad \text{--- (1)}$$

$$(3,6,6) = (k_1, k_1, 0) + (0, 2k_2, 3k_2) + (k_3, 2k_3, 3k_3)$$

$$(3,6,6) = (k_1 + 0 + k_3,\ k_1 + 2k_2 + 2k_3,\ 0 + 3k_2 + 3k_3)$$

Then system of equations is

$$k_1 + k_3 = 3$$

$$k_1 + 2k_2 + 2k_3 = 6$$

$$3k_2 + 3k_3 = 6$$

In matrix form

$$\begin{bmatrix} 1 & 0 & 1 & 3 \\ 1 & 2 & 2 & 6 \\ 0 & 3 & 3 & 6 \end{bmatrix}$$

$$R_2 - R_1,\ R_3 \times (1/3)$$

$$\begin{bmatrix} 1 & 0 & 1 & 3 \\ 0 & 2 & 1 & 3 \\ 0 & 1 & 1 & 2 \end{bmatrix}$$

$R_2 - R_3$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 \end{array}\right]$$

$R_3 - R_2$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$R_1 - R_3$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$K_1 = 2$ , $K_2 = 1$ , $K_3 = 1$

Therefore ① becomes

$$(3,6,6) = 2(1,1,0) + 1(0,2,3) + 1(1,2,3)$$

Q2 (c)

$$(a, b, c) = K_1\vec{u}_1 + K_2\vec{u}_2 + K_3\vec{u}_3 + K_4\vec{u}_4$$

$$(a, b, c) = K_1(1,1,0) + K_2(0,2,3) + K_3(1,2,3) + K_4(3, 6,6).$$

$$(K_1, K_1, 0) + (0, 2K_2, 3K_2) + (K_3, 2K_3, 3K_3) + (3K_4, 6K_4, 6K_4) = (a, b, c)$$

Then system of equations is

$$K_1 + K_3 + 3K_4 = a$$

$$K_1 + 2K_2 + 2K_3 + 6K_4 = b$$

$$3K_2 + 3K_3 + 6K_4 = c$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & a \\ 1 & 2 & 2 & 6 & b \\ 0 & 3 & 3 & 6 & c \end{array}\right]$$

$R_2 - R_1$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & a \\ 0 & 2 & 1 & 3 & b-a \\ 0 & 3 & 3 & 6 & c \end{array}\right]$$

$-R_2 + R_3 \quad , \quad R_3 \times \left(\frac{1}{3}\right)$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & a \\ 0 & 1 & 2 & 3 & a-b+c \\ 0 & 1 & 1 & 2 & c/3 \end{array}\right]$$

$R_3 - R_2$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & a \\ 0 & 1 & 2 & 3 & a-b+c \\ 0 & 0 & -1 & -1 & -a+b-\frac{2}{3}c \end{array}\right]$$

$R_3 \times (-1)$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 3 & a \\ 0 & 1 & 2 & 3 & a-b+c \\ 0 & 0 & 1 & 1 & a-b+\frac{2}{3}c \end{array}\right]$$

$R_1 - R_3 \quad , \quad R_2 - 2R_3$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 2 & b-\frac{2}{3}c \\ 0 & 1 & 0 & 1 & -a+b-\frac{1}{3}c \\ 0 & 0 & 1 & 1 & a-b+\frac{2}{3}c \end{array}\right]$$

This means that the given vector's $\vec{u}_1, \vec{u}_2, \vec{u}_3, \vec{u}_4$ span $\mathbb{R}_3$.

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## Q2(d)

### Can S be basis for $\mathbb{R}_3$?

Ans. Since in part (a) we have verified that the given vector's in set S are linearly dependent and in part (c) we have verified that the set S span $\mathbb{R}_3$. By definition of basis, The set S should be linearly independent, Hence S does not form basis for $\mathbb{R}_3$.

Q#3
(a) Given intersections are A, B, C, D

| Intersections | Flow in | Flow out | Equation Flow in = Flow out |
|---|---|---|---|
| A | $100+200$ | $x_1+x_4$ | $x_1+x_4=300$ |
| B | $x_1+x_2$ | $100+150$ | $x_1+x_2=250$ |
| C | $50+50$ | $x_2+x_3$ | $x_2+x_3=100$ |
| D | $x_3+x_4$ | $50+100$ | $x_3+x_4=150$ |

Hence system of equations becomes

$$1x_1+0x_2+0x_3+1x_4=300 \quad \text{—— (1)}$$
$$1x_1+1x_2+0x_3+0x_4=250 \quad \text{—— (2)}$$
$$0x_1+1x_2+1x_3+0x_4=100 \quad \text{—— (3)}$$
$$0x_1+0x_2+1x_3+1x_4=150 \quad \text{—— (4)}$$

(b) Gauss Jordan Method:

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 300 \\ 1 & 1 & 0 & 0 & 250 \\ 0 & 1 & 1 & 0 & 100 \\ 0 & 0 & 1 & 1 & 150 \end{array}\right]$$

$R_2 - R_1$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 300 \\ 0 & 1 & 0 & -1 & -50 \\ 0 & 1 & 1 & 0 & 100 \\ 0 & 0 & 1 & 1 & 150 \end{array}\right]$$

$R_3 - R_2$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 300 \\ 0 & 1 & 0 & -1 & -50 \\ 0 & 0 & 1 & 1 & 150 \\ 0 & 0 & 1 & 1 & 150 \end{array}\right]$$

$R_4 - R_3$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 300 \\ 0 & 1 & 0 & -1 & -50 \\ 0 & 0 & 1 & 1 & 150 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

By Back Substitution

$$x_3 + x_4 = 150$$

Let $x_4 = t$ ( Free variable)

Then

$$x_3 + t = 150$$

$$\boxed{x_3 = 150 - t}$$

From Row 2, we get

$$x_2 - x_4 = -50$$

$$x_2 - t = -50$$

$$\boxed{x_2 = t - 50}$$

From Row 1, we get

$$x_1 + x_4 = 300$$

$$x_1 + t = 300$$

$$\boxed{x_1 = 300 - t}$$

Hence

$$x_1 = 300 - t$$

$$x_2 = t - 50$$

$$x_3 = 150 - t$$

$$x_4 = t$$

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Q#3 (c)

Flow 1: Let $t = 100$

$$x_1 = 300 - 100 = 200$$

$$x_2 = 100 - 50 = 50$$

$$x_3 = 150 - 100 = 50$$

$$x_4 = 100$$

$$\Rightarrow (x_1, x_2, x_3, x_4) = (200, 50, 50, 100)$$

Flow 2: Let $t = 200$

$$x_1 = 300 - 200 = 100$$

$$x_2 = 200 - 50 = 150$$

$$x_3 = 150 - 200 = -50$$

$$x_4 = 200$$

$$\Rightarrow (x_1, x_2, x_3, x_4) = (100, 150, -50, 200)$$

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## Q#3 (d)

To find the minimum traffic flow along branch AB, we need to analyze

$$x_1 = 300 - t$$

Now, '$x_1$' is minimized when 't' is maximized. That is;

$$300 - t \geq 0$$
$$\Rightarrow 300 \geq t$$
$$\Rightarrow t \leq 300$$

Given this, maximum value of $t$ is $300$.

So

$$x_1 = 300 - 300 = 0$$

$$\boxed{x_1 = 0}$$

Thus, minimum traffic flow along the branch AB is zero.

Q#4 Given eq. of circle is

$$(x+2)^2 + (y-1)^2 = 4 \quad \text{— ①}$$

(a) Translation :

By definition ; we have

$$T(\vec{x}) = \vec{x} + \vec{a}$$

Then $\vec{x} = \begin{bmatrix} x \\ y \end{bmatrix}$ , and $\vec{a} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ (given)

So

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x+2 \\ y+3 \end{bmatrix}$$

$\Rightarrow \quad x' = x + 2 \quad ; \quad y' = y + 3$

$\Rightarrow \quad x' - 2 = x \quad ; \quad y' - 3 = y$

$\Rightarrow \quad \boxed{x = x' - 2} \quad ; \quad \boxed{y = y' - 3}$

putting values in eq ①, we have

$$(x' - \cancel{2} + \cancel{2})^2 + (y' - 3 - 1)^2 = 4$$

$$\boxed{(x')^2 + (y'-4)^2 = 4} \quad \text{— ②}$$

#4(b)

The translated equation obtained from part (a) is :

$$(x')^2 + (y'-4)^2 = 4 \quad \text{——} \; (2)$$

Now, stretching parallel to $y$-axis by a factor 3 :

Here $K = 3$ , and

$$A = \begin{bmatrix} 1 & 0 \\ 0 & K \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$$

By definition ;

$$T(\vec{x}) = A\vec{x}$$

$$T\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} x' \\ y' \end{bmatrix}$$

$$\begin{bmatrix} x'' \\ y'' \end{bmatrix} = \begin{bmatrix} x' \\ 3y' \end{bmatrix}$$

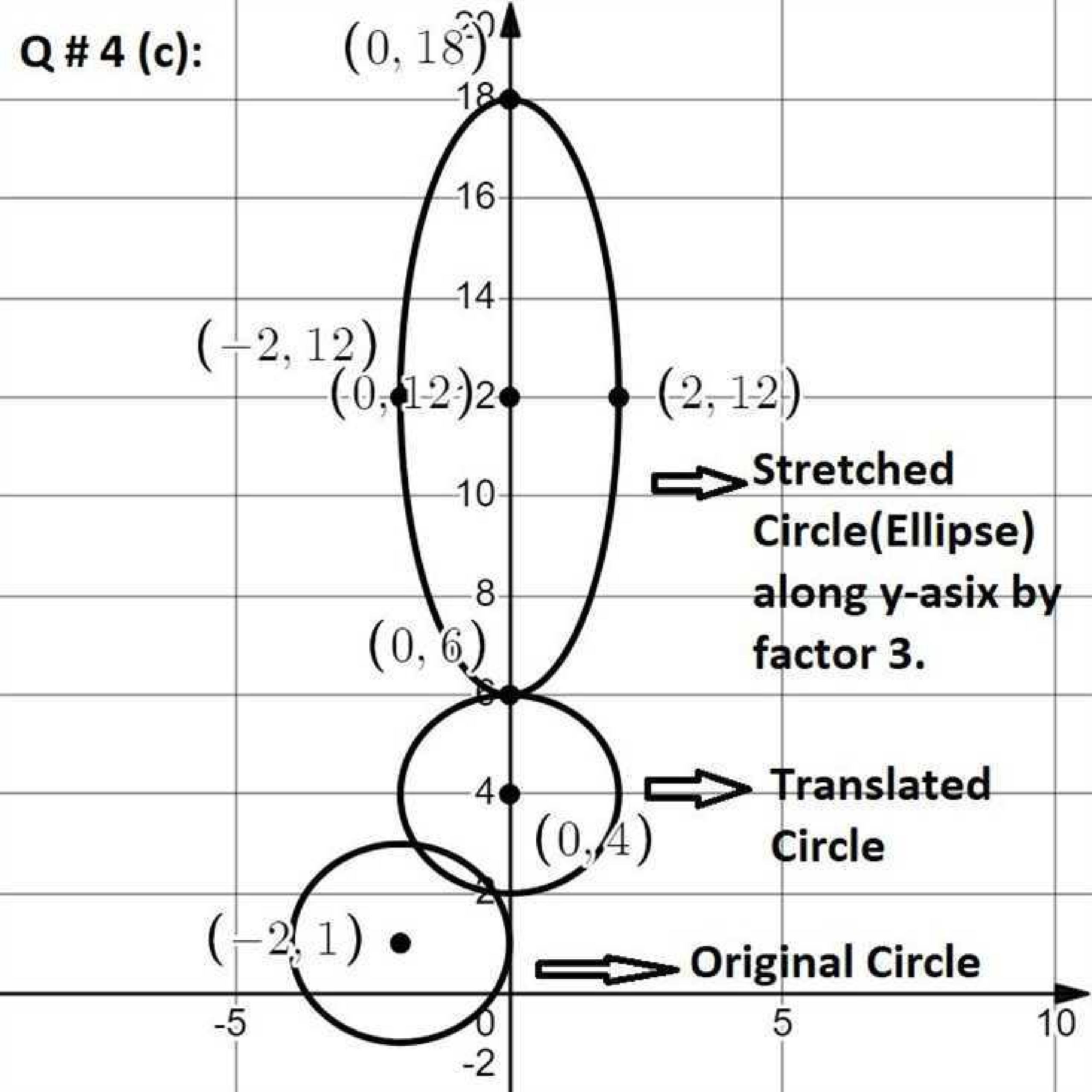
$$\Rightarrow \quad x'' = x' \quad \Rightarrow \boxed{x' = x''}$$

$$\Rightarrow \quad y'' = 3y' \quad \Rightarrow \boxed{y' = \tfrac{1}{3}y''}$$

putting values in eq (2), we get

$$(x'')^2 + \left(\tfrac{1}{3}y'' - 4\right)^2 = 4 \quad \text{——} \; (3)$$

**Q # 4 (c):**



$(0, 18)$
$(-2, 12)$
$(0, 12)$
$(2, 12)$
Stretched Circle(Ellipse) along y-asix by factor 3.
$(0, 6)$
Translated Circle
$(0, 4)$
$(-2, 1)$
Original Circle
20
18
16
14
12
10
8
6
4
2
0
-2
-5
5
10

Q#5 (a) Encryption:

Plain Text: SECRET WORLD

$$K = \begin{bmatrix} 5 & 1 \\ 2 & 7 \end{bmatrix}$$

Sol: From Table

| | | | |
|---|---|---|---|
| $S \longrightarrow 19$ | | $W \longrightarrow 23$ | |
| $E \longrightarrow 5$ | | $O \longrightarrow 15$ | |
| $C \longrightarrow 3$ | | $R \longrightarrow 18$ | |
| $R \longrightarrow 18$ | | $L \longrightarrow 12$ | |
| $E \longrightarrow 5$ | | $D \longrightarrow 4$ | |
| $T \longrightarrow 20$ | | $D \longrightarrow 4$ (Dummy) | |

By definition

$$C = KP \pmod{26}$$

$$C = \underbrace{\begin{bmatrix} 5 & 1 \\ 2 & 7 \end{bmatrix}}_{2\times2} \begin{bmatrix} 19 & 3 & 5 & 23 & 18 & 4 \\ 5 & 18 & 20 & 15 & 12 & 4 \end{bmatrix}_{2\times6} \pmod{26}$$

$$C = \begin{bmatrix} 100 & 33 & 45 & 130 & 102 & 24 \\ 73 & 132 & 150 & 151 & 120 & 36 \end{bmatrix} \pmod{26}.$$

$$C = \begin{bmatrix} 22 & 7 & 19 & 26 & 24 & 24 \\ 21 & 2 & 20 & 21 & 16 & 10 \end{bmatrix} \bmod 26)$$

$$
\begin{array}{cccccc|cccccc}
22 & 21 & 7 & 2 & 19 & 20 & 26 & 21 & 24 & 16 & 24 & 10 \\
\downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\
V & U & G & B & S & T & Z & U & X & P & X & J \\
 & & & & & & & & & & & \downarrow \\
 & & & & & & & & & & & \text{(Dummy)}
\end{array}
$$

Hence ciphered text is

VUGBST ZUXPX

(b) Inverse of K :

$$|K| = \begin{vmatrix} 5 & 1 \\ 2 & 7 \end{vmatrix} = 5(7) - 2(1) = 35 - 2 = 33$$

$$\boxed{|K| = 7 \pmod{26}}$$

$$Adj(K) = \begin{bmatrix} 7 & -1 \\ -2 & 5 \end{bmatrix}$$

Now let

$$x = \frac{1}{7} \pmod{26}$$

$$\Rightarrow 7x = 1 \pmod{26}$$

$$\Rightarrow 7(15) = 1 \pmod{26}$$

$$\Rightarrow 7^{-1} = 15$$

Formula:

$$K^{-1} = \frac{1}{|K|} Adj(K) = \frac{1}{7}\begin{bmatrix} 7 & -1 \\ -2 & 5 \end{bmatrix} \pmod{26}$$

$$K^{-1} = 15\begin{bmatrix} 7 & -1 \\ -2 & 5 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 105 & -15 \\ -30 & 75 \end{bmatrix} \pmod{26}$$

$$\boxed{K^{-1} = \begin{bmatrix} 1 & 11 \\ 22 & 23 \end{bmatrix} \pmod{26}}$$